LAB#12
User Level Security

How to Create a New User

Introduction

MySQL is an open-source database management software that helps users store, organize, and later retrieve data.

How to Create a New User

Let’s start by making a new user within the MySQL shell:

- CREATE USER 'newuser'@'localhost' IDENTIFIED BY 'password';

At this point newuser has no permissions to do anything with the databases. In fact, even if newuser tries to login (with the password, password), they will not be able to reach the MySQL shell.

Therefore, the first thing to do is to provide the user with access to the information they will need.

- GRANT ALL PRIVILEGES ON * . * TO 'newuser'@'localhost';

The asterisks in this command refer to the database and table (respectively) that they can access—this specific command allows to the user to read, edit, execute and perform all tasks across all the databases and tables.

Once you have finalized the permissions that you want to set up for your new users, always be sure to reload all the privileges.

- FLUSH PRIVILEGES;

Copy Your changes will now be in effect.

How To Grant Different User Permissions

Here is a short list of other common possible permissions that users can enjoy.

- ALL PRIVILEGES- as we saw previously, this would allow a MySQL user full access to a designated database (or if no database is selected, global access across the system)
- CREATE- allows them to create new tables or databases
- DROP- allows them to them to delete tables or databases
- DELETE- allows them to delete rows from tables
- INSERT- allows them to insert rows into tables
- SELECT- allows them to use the SELECT command to read through databases
- UPDATE- allow them to update table rows
- GRANT OPTION- allows them to grant or remove other users’ privileges

To provide a specific user with a permission, you can use this framework:

- GRANT type_of_permission ON database_name.table_name TO 'username'@'localhost';

If you want to give them access to any database or to any table, make sure to put an asterisk (*) in the place of the database name or table name.

Each time you update or change a permission be sure to use the Flush Privileges command.

If you need to revoke a permission, the structure is almost identical to granting it:

- REVOKE type_of_permission ON database_name.table_name FROM 'username'@'localhost';

Note that when revoking permissions, the syntax requires that you use FROM, instead of TO as we used when granting permissions.

You can review a user's current permissions by running the following:

- SHOW GRANTS FOR 'username'@'localhost';

Just as you can delete databases with DROP, you can use DROP to delete a user altogether:

- DROP USER 'username'@'localhost';

To test out your new user, log out by typing:

- quit
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Copy and log back in with this command in terminal:

- mysql -u [username] -p

What is ER Diagram?

ER Diagram stands for Entity Relationship Diagram, also known as ERD is a diagram that displays the relationship of entity sets stored in a database.

At first look, an ER diagram looks very similar to the flowchart. However, ER Diagram includes many specialized symbols, and its meanings make this model unique. The purpose of ER Diagram is to represent the entity framework infrastructure.